

Curvilinear Geometry, Primary Structure, and Gröbner Degenerations of Conductor Fiber Cones in a Huneke–Wiegand Family

Felipe Santibañez-Leal 

CAOS Research programme, github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

For every integer $p \geq 4$, a previously constructed symmetric numerical semigroup ring R_p and rigid two-generated monomial ideal determine a conductor ideal T_p whose special fiber $\mathcal{C}_p = \mathcal{F}(T_p)$ is one-dimensional Cohen–Macaulay, generated in degree one by $10p$ monomials, and has multiplicity $24p$. We prove that this fiber cone admits the explicit standard-graded parametrization

$$\mathcal{C}_p \cong \mathbb{k}[xy^a : a \in G_p] \subseteq \mathbb{k}[x, y]/(y^{24p}), \quad \deg(xy^a) = 1,$$

where G_p is the exact degree-one offset set. If $P_p = \mathbb{k}[X_a : a \in G_p]$ and J_p is the defining ideal, then

$$\sqrt{J_p} = L_p = (X_a : a > 0), \quad J_p \text{ is } L_p\text{-primary.}$$

Thus the complete primary decomposition has one component. The nilradical of $\mathcal{C}_p$ has sharp nilpotency index $24p$: its $(24p - 1)$ st power is nonzero, while its $24p$ th power vanishes. Dehomogenization at X_0 gives

$$\mathcal{C}_p/(X_0 - 1) \cong \mathbb{k}[y]/(y^{24p}).$$

Consequently $\text{Proj}(\mathcal{C}_p)$ is a saturated length- $24p$ curvilinear fat point with one-dimensional Zariski tangent space. It is locally Gorenstein, although its homogeneous coordinate ring has Cohen–Macaulay type $10p + 1$ and is neither level nor Gorenstein. We also compute its affine Kähler differential module, including the exact characteristic split. The proof is deductive from previously proved offset-basis and Cohen–Macaulay input theorems. We further give the complete reduced Gröbner basis for graded reverse lexicographic order with X_0 last. Its degree profile is

$$(50p^2 - 17p, 5p - 1, p - 2)$$

in degrees two, three, and four, with no later elements. No minimal leading monomial contains X_0 , yielding a flat Cohen–Macaulay monomial degeneration. Exact parameter sweeps, an independently encoded clique-based recomputation, and an all-parameter Presburger boundary certificate validate the formulas and run negative controls, but do not replace the deductive arguments.

1 Introduction

Fiber cones encode the number and relations of generators of powers of an ideal. Even for monomial ideals, their reducedness and projective geometry can vary sharply. Explicit defining ideals are known for several convex, concave, symmetric, and low-generator classes [4, 5]; in particular, nonradical monomial fiber cones occur naturally. Curvilinear zero-dimensional schemes provide a complementary geometric language: their local maximal ideals are generated by one element, and their differential modules give effective signatures of this property [6].

This paper brings those two viewpoints together for the conductor fiber cones attached to an explicit infinite family of Huneke–Wiegand counterexamples. The family was constructed in [3] after Son Pham’s discovery of a first public counterexample [2], which Craig Huneke independently verified; the discovery priority and the present contribution are distinct. The same earlier work determines the conductor fiber cone, its complete degreewise offset bases, its Cohen–Macaulay type, its defining ideal, and exact edges of its graded resolution. Here we use that information from a different angle. The presence of offsets 0 and 1, combined with a hard truncation at $24p$, makes the entire projective scheme one-variable after dehomogenization.

The main result is stronger than a radical calculation. It gives a concrete truncated monomial model, a complete one-component primary decomposition, the exact nilpotency exponent, and the scheme-theoretic affine chart. It also separates two notions that are easy to conflate: the projective scheme is locally Gorenstein, while its given homogeneous embedding is not arithmetically Gorenstein.

Throughout, derived theorem statements are marked [D] by convention, while finite exact computations and solver outcomes are machine-verified [MV]. No [MV] result is used as an extrapolation principle.

2 The conductor family and imported input

Fix $p \geq 4$ and put $s = 6p$ and $q = 4s = 24p$. Define residue sets in $[0, s - 1]$ by

$$\begin{aligned} A_p &= [0, p] \cup [3p, 4p - 2], \\ B_p &= ([p + 1, 3p - 1] \setminus \{2p - 1\}) \cup \{4p\} \cup [5p - 1, 6p - 1], \\ D_p &= [0, 2p] \cup [3p, 5p - 2]. \end{aligned}$$

The numerical semigroup Γ_p contains

$$\{0\}, \quad 4s + A_p, \quad [5s, 6s - 1], \quad 6s + B_p, \quad 8s + D_p, \quad [9s, 13s - 2], \quad [13s, \infty),$$

and no other nonnegative integers. The earlier family theorem proves that Γ_p is symmetric, with multiplicity $24p$ and Frobenius number $78p - 1$, and that $I_p = (t^{24p}, t^{30p})$ is a nonprincipal rigid ideal in the localization of $\mathbb{k}[t^{\Gamma_p}]$ at its positive-degree maximal ideal [3].

Let $E_p = \text{End}_{R_p}(I_p)$ and let $T_p = R_p : E_p$ be the conductor. The object studied here is its special fiber

$$\mathcal{C}_p = \mathcal{F}(T_p) = \bigoplus_{n \geq 0} T_p^n / \mathfrak{m}_p T_p^n.$$

We recall the exact input in the form needed below.

Proposition 2.1 (Imported offset-basis input). *There is a presentation*

$$P_p = \mathbb{k}[X_a : a \in G_p] \longrightarrow \mathcal{C}_p, \quad X_a \longmapsto [t^{q+a}],$$

where $|G_p| = 10p$ and

$$G_p = \{0\} \cup [1, p] \cup [3p, 4p - 2] \cup [6p, 8p - 2] \cup [8p, 10p - 2] \cup \{10p\} \\ \cup [11p - 1, 12p - 1] \cup [13p + 1, 14p - 2] \cup [14p, 15p - 1] \cup \{16p\} \cup [17p - 1, 18p - 1].$$

If J_p is the kernel, a degree- n monomial is zero precisely when its total offset is absent from E_n , and two nonzero monomials have the same class precisely when their total offsets agree. The cumulative bases are

$$\begin{aligned} E_1 &= G_p, \\ E_2 &= [0, 2p] \cup [3p, 5p - 2] \cup [6p, q - 1], \\ E_3 &= [0, q - 1] \setminus \{6p - 1\}, \\ E_n &= [0, q - 1] \quad (n \geq 4). \end{aligned}$$

Moreover, $\mathcal{C}_p$ is a one-dimensional Cohen–Macaulay standard graded algebra, free of rank q over $\mathbb{k}[X_0]$, and has Cohen–Macaulay type $10p + 1$. Its Artinian reduction by X_0 has socle in two degrees, so $\mathcal{C}_p$ is neither level nor Gorenstein.

Remark 2.2. Proposition 2.1 packages the previously proved conductor, fiber-cone, presentation, and offset-congruence theorems. The present paper does not infer it from the finite computations below. Its exact all-parameter component step retains the solver/encoding boundary disclosed in [3].

3 The truncated monomial model

Assign standard degree one to each monomial xy^a and define

$$\mathcal{A}_p = \mathbb{k}[xy^a : a \in G_p] \subseteq \mathbb{k}[x, y]/(y^q).$$

Theorem 3.1 (Truncated parametrization). *For every $p \geq 4$, the homomorphism*

$$P_p \longrightarrow \mathcal{A}_p, \quad X_a \longmapsto xy^a,$$

induces a standard-graded isomorphism $\mathcal{C}_p \cong \mathcal{A}_p$.

Proof. A degree- n monomial with total offset b maps to $x^n y^b$. This is zero exactly when $b \geq q$, and distinct surviving offsets give linearly independent monomials. By Proposition 2.1, the corresponding class in $(\mathcal{C}_p)_n$ is zero precisely when the offset is absent from E_n , and otherwise depends only on the offset. The achieved surviving offsets are exactly E_n in both quotients. Hence the two kernels agree in every graded degree. $\square$

Corollary 3.2 (Exact dehomogenization). *There is an isomorphism*

$$\mathcal{C}_p/(X_0 - 1) \cong \mathbb{k}[y]/(y^q), \quad X_a \longmapsto y^a.$$

Proof. The offsets 0 and 1 belong to G_p . For every positive $a \in G_p$, the two degree- a monomials $X_0^{a-1}X_a$ and X_1^a have the same surviving offset $a < q$, so

$$X_0^{a-1}X_a = X_1^a.$$

After setting $X_0 = 1$, every generator is a power of X_1 , and $X_1^q = 0$. This gives a surjection from $\mathbb{k}[y]/(y^q)$. Freeness of rank q over $\mathbb{k}[X_0]$ shows that the target also has vector-space dimension q , hence the surjection is an isomorphism. $\square$

4 Primary decomposition and sharp nilpotence

Let

$$L_p = (X_a : a > 0) \subseteq P_p, \quad N_p = L_p/J_p \subseteq \mathcal{C}_p.$$

Theorem 4.1 (One primary component). *For every $p \geq 4$,*

$$\sqrt{J_p} = L_p, \quad J_p \text{ is } L_p\text{-primary.}$$

Thus J_p itself is the complete primary decomposition. The nilradical N_p has exact nilpotency index $q = 24p$:

$$N_p^{q-1} \neq 0, \quad N_p^q = 0.$$

Proof. The quotient $\mathcal{C}_p/N_p$ is $\mathbb{k}[X_0]$, so it is reduced. Every product of q generators with positive offset has total offset at least q and vanishes by Theorem 3.1. Hence N_p is nilpotent and equals the nilradical. It is the unique minimal prime. The element X_1^{q-1} maps to $x^{q-1}y^{q-1}$ and is nonzero, proving the sharp exponent.

The unique-minimal-prime statement alone would not prove primaryness. Here the Cohen–Macaulay hypothesis is essential: a one-dimensional Cohen–Macaulay ring has no embedded associated primes. Therefore $\text{Ass}(\mathcal{C}_p) = \{N_p\}$, so the zero ideal is N_p -primary. Taking its inverse image in P_p proves the assertion about J_p . $\square$

Remark 4.2 (Why Cohen–Macaulayness is load-bearing). The ring $\mathbb{k}[u, v]/(u^2, uv)$ has a unique minimal prime but also an embedded associated prime; its zero ideal is not primary. This example shows that the associated-prime argument cannot be replaced by a radical-only inference.

5 Curvilinear geometry and Gorenstein separation

Theorem 5.1 (The projective fat point). *The ideal J_p is saturated for the irrelevant ideal. The projective scheme $\text{Proj}(\mathcal{C}_p)$ is supported at $[1 : 0 : \cdots : 0]$ and is a length- q curvilinear fat point. Its Zariski tangent space has dimension one. The scheme is locally Gorenstein, while the homogeneous coordinate ring $\mathcal{C}_p$ is not level and not Gorenstein.*

Proof. Because $\mathcal{C}_p$ is one-dimensional Cohen–Macaulay, its zeroth local cohomology for the irrelevant ideal vanishes. The identity $J_p^{\text{sat}}/J_p = H_{P_{p,+}}^0(\mathcal{C}_p)$ gives saturation. Theorem 4.1 shows that the support is the single coordinate point and hence lies entirely in $D_+(X_0)$. Corollary 3.2 identifies its affine coordinate ring with $\mathbb{k}[y]/(y^q)$. This has length q and principal maximal ideal (y) , so the scheme is curvilinear and its tangent space $(y)/(y^2)$ is one-dimensional. Its socle is generated by y^{q-1} , proving local Gorensteinness. Proposition 2.1 gives type $10p+1$ and nonlevelness for the homogeneous coordinate ring, proving the arithmetic contrast. $\square$

Remark 5.2. This example makes the embedding distinction concrete:

$$\text{locally Gorenstein} \not\Rightarrow \text{arithmetically Gorenstein.}$$

The local scheme is always the one-variable truncation, while the standard grading retains the nonlevel offset-layer structure.

6 Kähler differentials

The curvilinear chart has a short differential fingerprint which also records the ground-field characteristic.

Proposition 6.1 (Differential signature). *Let $B_p = \mathcal{C}_p/(X_0 - 1)$. Then*

$$\Omega_{B_p/\mathbb{k}}^1 \cong (B_p/(qy^{q-1})) dy, \quad \bigwedge^m \Omega_{B_p/\mathbb{k}}^1 = 0 \quad (m \geq 2).$$

Consequently,

$$\dim_{\mathbb{k}} \Omega_{B_p/\mathbb{k}}^1 = \begin{cases} q-1, & \text{char}(\mathbb{k}) \nmid q, \\ q, & \text{char}(\mathbb{k}) \mid q. \end{cases}$$

The cotangent space at the closed point has dimension one in every characteristic.

Proof. Apply the conormal sequence to $B_p = \mathbb{k}[y]/(y^q)$. Differentiating the defining equation gives $qy^{q-1}dy = 0$, which proves the presentation and the characteristic split. The differential module is cyclic, so every exterior power of degree at least two vanishes. Tensoring with $B_p/(y)$ leaves the single generator dy . $\square$

This agrees with the general differential characterization of weakly curvilinear schemes in [6], while the exact truncation and characteristic formula here come from the conductor-family parametrization.

7 The canonical grevlex degeneration

Order the variables by decreasing offset and use graded reverse lexicographic order $\prec$; thus $X_a \succ X_b$ exactly when $a > b$, and X_0 is last. If a monomial is written as a nondecreasing tuple of offsets, then the $\prec$ -smallest monomial in a fixed degree and offset fiber is the lexicographically smallest tuple. Write it as $N_{n,b}$. This last-parameter order and the use of finite standard-monomial counts are consistent with Gröbner methods for projective monomial curves and associated graded semigroup algebras [7, 8]; the staircase and formulas below are specific to the present truncated conductor family.

Lemma 7.1 (Canonical staircase). *In each surviving bidegree (n, b) , the unique standard monomial of $P_p/\text{in}_{\prec}(J_p)$ is $N_{n,b}$. A monomial is a minimal generator of the initial ideal exactly when it is noncanonical and every one-variable divisor is canonical.*

Proof. Theorem 3.1 shows that two nonzero degree- n monomials agree precisely when their total offsets agree, and that all monomials of offset at least q vanish. Every surviving fiber therefore contributes its unique $\prec$ -smallest monomial to the standard basis. The last assertion is the minimal-boundary criterion for a monomial ideal. $\square$

Theorem 7.2 (Explicit reduced Gröbner basis). *For every $p \geq 4$, the reduced Gröbner basis of J_p has degree profile*

$$(50p^2 - 17p, 5p - 1, p - 2)$$

in degrees two, three, and four. There are no basis elements in degree at least five.

The quadratic part contains one relation for every noncanonical quadratic monomial M of total offset b :

$$M - N_{2,b} \quad (b < q), \quad M \quad (b \geq q).$$

Among these, $(77p^2 - 49p + 2)/2$ are binomials and $(23p^2 + 15p - 2)/2$ are monomial zero-relations. The cubic leading monomials are

$$\begin{aligned} A_i &= X_i X_p X_{12p-1} & (1 \leq i \leq p), \\ B_i &= X_i X_p X_{15p-1} & (1 \leq i \leq p), \\ C_i &= X_i X_p X_{18p-1} & (1 \leq i \leq p), \\ D_i &= X_i X_{4p-2} X_{16p} & (1 \leq i \leq p-2), \\ E_i &= X_i X_{4p-2} X_{18p-1} & (1 \leq i \leq p), \end{aligned}$$

together with X_p^3 . Their reduced tails are

$$\begin{aligned} A_1 &\mapsto X_0 X_{3p} X_{10p}, & A_i &\mapsto X_0^2 X_{13p+i-1} & (2 \leq i \leq p-1), \\ A_p &\mapsto X_0 X_1 X_{14p-2}, & B_i &\mapsto X_0 X_{i-1} X_{16p} & (1 \leq i \leq p-1), \\ B_p &\mapsto X_0^2 X_{17p-1}, & C_i &\mapsto X_0 X_{3p+i-1} X_{16p} & (1 \leq i \leq p-1), \\ C_p &\mapsto X_0 X_{3p} X_{17p-1}, & D_i &\mapsto X_0 X_{3p} X_{17p+i-2}, \\ E_i &\mapsto X_0 X_{7p+i-2} X_{15p-1} & (i = 1, 2), \\ E_i &\mapsto X_0 X_{6p+i-3} X_{16p} & (3 \leq i \leq p), \\ X_p^3 &\mapsto X_0^2 X_{3p}. \end{aligned}$$

Finally, the quartics are

$$X_i X_p^2 X_{4p-2} - X_0^3 X_{6p+i-2}, \quad 2 \leq i \leq p-1.$$

No minimal leading monomial is divisible by X_0 .

Proof. There are $\binom{10p+1}{2} = 50p^2 + 5p$ quadratic monomials and $|E_2| = 22p$ canonical ones, giving $50p^2 - 17p$ quadratic boundary elements. Summing the unordered pairs from the eleven blocks of G_p whose offset is at least q gives $(23p^2 + 15p - 2)/2$ zero-relations; subtraction gives the stated binomial count.

For the nonquadratic boundary, Lemma 7.1 turns completeness into Presburger arithmetic over the eleven interval blocks of G_p . The exact certificate proves, for all integers $p \geq 4$, that the five indexed cubic groups plus X_p^3 are both sound and complete, that the displayed quartic family is sound and complete, and that all twelve endpoint/interior tail cases are canonical and have the correct offset. Its 16 negated obligations are UNSAT. The family sizes give $p+p+p+(p-2)+p+1 = 5p-1$ cubics and $p-2$ quartics.

It remains to close the infinite degree tail without solver extrapolation. Since $E_4 = [0, q-1]$ and $E_n = [0, q-1]$ for $n \geq 4$, the last-variable order forces

$$N_{n,b} = X_0^{n-4} N_{4,b} \quad (n \geq 4, 0 \leq b < q).$$

The staircase therefore stabilizes in degree four, so no later minimal boundary can occur. The listed polynomials are monic with distinct minimal leading monomials and canonical tails; hence they form the reduced Gröbner basis. $\square$

Corollary 7.3 (Flat Cohen–Macaulay degeneration). *The initial algebra is Cohen–Macaulay, X_0 is regular on it, and its Artinian reduction has Hilbert function*

$$(1, 10p-1, 12p, 2p-1, 1).$$

The defining ideal has relation type three, while this reduced Gröbner basis reaches degree four.

Proof. No minimal initial generator contains X_0 , so multiplication by X_0 is injective on standard monomials. The successive offset layers from Proposition 2.1 have the displayed sizes. Their sum is $24p$, and adjoining powers of X_0 recovers the stabilized staircase. Relation type three was established by the earlier minimal-presentation theorem, whereas Theorem 7.2 has $p - 2 > 0$ quartics. $\square$

8 Computational verification

The symbolic proofs above use the imported results in Proposition 2.1. Computation is used to validate their translation into the new parametrization and to exercise negative controls. The primary-structure sweep computes truncated sumsets directly from G_p for every $p = 4, \dots, 300$. An independent recomputation starts from the five disjoint Artinian layers. The Gröbner sweep independently checks all families listed in Theorem 7.2, canonical tails, divisor minimality, Hilbert values, and controls at the same 297 parameters. A separately encoded clique-based recomputation enumerates canonical pairs and reconstructs factorization complexes through degree six at $p = 4, 5, 6, 17, 73, 151, 300$.

computation	coverage	SHA-256 aggregate
primary sweep	all $p = 4, \dots, 300$	f3373f4f58287fd3...f9d6b39b6
primary recomputation	all rows; six detailed rebuilds	84c00be8ff64002...f5b6c8f143e
Gröbner sweep	all $p = 4, \dots, 300$	63af8f734afc8c05...9e4365a218
clique recomputation	seven detailed rebuilds	401c4807cc0a29a6...586766e41373
Presburger boundary	all integers $p \geq 4$	10c66bbcaa56108f...e046f9dd5fa

The negative controls delete offset 1, perturb the truncation and nilpotency exponents, corrupt the degree-three hole, omit a radical coordinate, conflate local and arithmetic Gorensteinness, erase the differential characteristic split, and remove the no-embedded-primes premise. Every mutation is rejected. Exact scripts, artifacts, premise hashes, and proofs are archived in the public repository as the computation records `EXP-025-curvilinear-primary-structure` and `EXP-026-grevlex-staircase`, in the folder `problems/commutative-algebra/huneke-wiegand/experiments/`.

The finite sweep does not prove the theorem for all p . The infinite conclusion is supplied by Theorems 3.1–5.1 and Proposition 6.1. A defect in an imported input theorem would propagate even if both implementations agreed. The Presburger boundary certificate uses Z3 4.16.0 and does not emit independently checked UNSAT proof objects; the canonical-staircase and stabilization arguments remain separate deductive steps.

9 Scope and further questions

The result concerns the conductor fiber cones of the explicit family recalled in Section 2. It does not assert that arbitrary monomial fiber cones are primary, curvilinear, Cohen–Macaulay, or locally Gorenstein. It does not strengthen the underlying Huneke–Wiegand counterexample theorem for arbitrary modules or arbitrary one-dimensional Gorenstein domains.

With the primary decomposition and the reduced Gröbner basis determined, one major algebraic question remains open for this family:

1. Can the unresolved interior of the graded Betti table be expressed by closed formulas, or does it exhibit parameter-dependent combinatorial phases?

The monomial degeneration may help by replacing value-semigroup products with the combinatorics of one explicit finite staircase.

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Declaration of generative AI use. Generative AI tools were used in the preparation of this paper. The author reviewed and verified all content and takes full responsibility for the work.

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